

Pricing and Hedging a Structured Floating Rate Note

SMM269 Fixed Income — Group 9

Bayes Business School, City St George's, University of London

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Camilla Ayres [250061774] Renzo Gonzales Delgado [250046860]
Pranay Gudipati [250044463] Jia Yi Valentina Wang [220057189]

1 Introduction

This report values and risk-manages a structured floating-rate note issued by UniCredit S.p.A. (IT0005599110). As of the trade date 5 November 2025 (spot 7 November 2025), the note pays quarterly coupons linked to 3-month EURIBOR with a 160% participation factor, a 0% floor and a 5.45% cap, and redeems at par on 12 June 2034. Valuation requires decomposing the payoff into standard instruments, modelling embedded optionality, and incorporating issuer credit risk.

2 Product Description and Market Data

Table 1: Key contractual terms. ISIN IT0005599110, UniCredit S.p.A.

Issuer	UniCredit S.p.A.	Participation	160%
ISIN	IT0005599110	Cap	5.45% p.a.
Maturity	12 Jun 2034 (10Y)	Floor	0.00% p.a.
Denomination	€1,000	Frequency	Quarterly
Reference	3m EURIBOR (EUR003M)	Day count	30/360 (Bond Basis)
Reset	−2 TARGET days before period start	BDC	Following (Unadjusted)
Redemption	At par (100%)	Settlement	T+2

Modelling assumption. The MREL call option (Final Terms, paragraph 20) is assumed not to be exercised; the bond is modelled as outstanding until 12 June 2034.

2.1 Bond Structure and Coupon Mechanics

Let L_i denote the 3-month EURIBOR fixing on the Interest Determination Date for period i (second TARGET business day before period start). The annualised coupon rate is

$$R(L_i) = \min(0.0545, \max(0, 1.6L_i)), \quad (1)$$

and the cash flow is $\text{Coupon}_i = N R(L_i) \alpha_i$, $N = €1,000$, $\alpha_i = 0.25$ (30/360). The cap binds at $L^* = 5.45\%/1.6 = 3.40625\%$, giving:

$$R(L) = \begin{cases} 0, & L \leq 0, \\ 1.6L, & 0 < L < L^*, \\ 5.45\%, & L \geq L^*. \end{cases}$$

Coupon replication. Applying the collar identity to (1):

$$R(L_i) = \underbrace{1.6 L_i}_{\text{(i) leveraged float}} + \underbrace{1.6(-L_i)^+}_{\text{(ii) long floorlet at 0\%}} - \underbrace{1.6(L_i - L^*)^+}_{\text{(iii) short caplet at } L^*=3.406\%} \quad (2)$$

Component (i) is a plain floating coupon at 160% of EURIBOR. Component (ii) is a long floorlet at strike zero: it pays $1.6 \max(-L_i, 0)$ when EURIBOR is negative, preventing the coupon from falling below zero. Component (iii) is a short caplet at L^* : it pays away $1.6(L_i - L^*)^+$ whenever EURIBOR exceeds the breakeven, capping the coupon at 5.45%. The replication is **static**. Extending to all

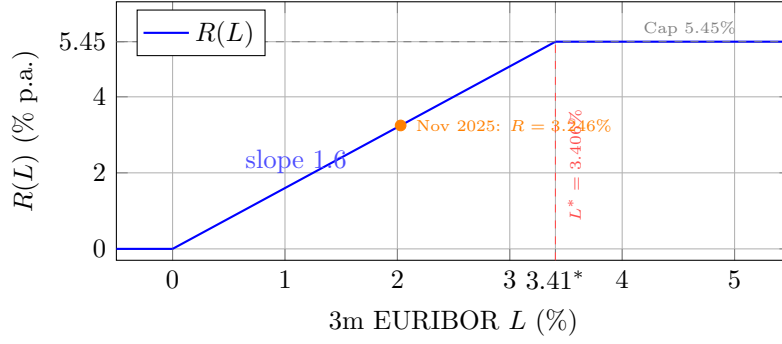


Figure 1: Coupon $R(L)$ vs 3m EURIBOR: floor ($L \leq 0$), linear region ($0 < L < 3.406\%$, slope 1.6), cap ($L \geq 3.406\%$, $R = 5.45\%$). Orange dot: Nov 2025 fixing $L_6 = 2.029\%$, $R_6 = 3.246\%$.

remaining periods and adding discounted par redemption:

$$V = N P(t_0, T_{40}) + \sum_{i=6}^{40} N \cdot 1.6 F_i \alpha_i P(t_0, T_i) + 1.6 N \sum_{i=7}^{40} [\text{Floorlet}_i(0\%) - \text{Caplet}_i(L^*)], \quad (3)$$

where $P(t_0, T_i)$ is the risk-free discount factor, F_i is the ACT/360 implied forward 3-month EURIBOR for period i , and α_i is the ACT/360 accrual fraction. Period 6 is already fixed at $L_6 = 2.029\%$ and contributes only to the FRN leg. The issuer monetises component (D): the premium collected

	Instrument	Role	Position
(A)	Zero-coupon bond	Par €1,000 at T_{40}	Long
(B)	Leveraged FRN ($\times 1.6$)	160% of 3m EURIBOR, periods 6–40	Long
(C)	34 floorlets at 0%	Prevents negative coupons, periods 7–40	Long $\times 1.6$
(D)	34 caplets at L^*	Caps coupon at 5.45%, periods 7–40	Short $\times 1.6$

Table 2: Static replication. Period 6 fixed; options on periods 7–40.

from selling the caplet strip offsets part of the funding cost, while the investor accepts the ceiling in exchange for the leveraged participation in the linear region. Component (C) provides downside insurance against negative EURIBOR.

2.2 Historical Coupons

EURIBOR fixings are from the Suomen Pankki EUR003M series [Suomen Pankki (Bank of Finland), 2025]. Table 3 shows the coupon path: the first two periods were capped at 5.45%, then EURIBOR fell below $L^* = 3.406\%$ and coupons moved into the linear region. Total paid: €56.742 vs €36.440 from a plain 100% FRN—a €20.302 premium from the 160% participation. The 0% floor was not activated historically but retains genuine option value, as EURIBOR traded negative from 2015 to 2022.

2.3 Market Data

2.3.1 Interest Rate Curve

The EUR discount curve is constructed from two sets of instruments. The short end uses five EURIBOR deposit rates (O/N through 6M), quoted on an Actual/360 basis. The medium and long end uses 17 EUR interest rate swap par rates (1Y through 50Y), with an annual fixed leg quoted on 30/360 (Bond Basis) versus a 6-month EURIBOR floating leg. All instruments are listed in Table 4. The O/N rate of 1.90% is broadly consistent with the ECB deposit facility rate as of November 2025. The curve is gently upward-sloping from 2Y ($\approx 2.14\%$) to 25Y ($\approx 2.91\%$) before inverting mildly at the ultra-long end (50Y: 2.83%), consistent with market expectations of gradual rate normalisation.

Reset	Start	End	Payment	3m EURIBOR	Coupon rate	Days [†]	Coupon (€)
10/06/24	12/06/24	12/09/24	12/09/24	3.7430%	5.4500%*	90	13.625
10/09/24	12/09/24	12/12/24	12/12/24	3.4600%	5.4500%*	90	13.625
10/12/24	12/12/24	12/03/25	12/03/25	2.8720%	4.5952%	90	11.488
10/03/25	12/03/25	12/06/25	12/06/25	2.5470%	4.0752%	90	10.188
10/06/25	12/06/25	12/09/25	12/09/25	1.9540%	3.1264%	90	7.816
10/09/25	12/09/25	12/12/25	12/12/25	2.0290%	3.2464% [†]	90	8.116 [†]
Total coupons paid (periods 1–5)							56.742
Total incl. current fixing (periods 1–6)							64.858

Table 3: Historical coupon fixings. *Cap binding: max €13.625 paid. [†]Fixed 10/09/25; payable 12/12/25; not yet paid at trade date. [‡]30/360 (90 days/quarter).

Table 4: EUR curve input instruments. Trade date 5 Nov 2025; spot date 7 Nov 2025. Business-day convention: Modified Following; calendar: TARGET.

Type	Tenor	Quote (%)	Type	Tenor	Quote (%)
EURIBOR	O/N	1.9000	IRS	6Y	2.4155
EURIBOR	1W	1.9210	IRS	7Y	2.4803
EURIBOR	1M	1.9140	IRS	8Y	2.5398
EURIBOR	3M	2.0230	IRS	9Y	2.5975
EURIBOR	6M	2.1420	IRS	10Y	2.6330
IRS	1Y	2.1990	IRS	15Y	2.8330
IRS	18M	2.1140	IRS	20Y	2.9040
IRS	2Y	2.1420	IRS	25Y	2.9090
IRS	3Y	2.1930	IRS	30Y	2.9080
IRS	4Y	2.2815	IRS	40Y	2.8825
IRS	5Y	2.3498	IRS	50Y	2.8295

2.3.2 Caplet Volatility Surface

Displaced Black VCAP3A EUR caplet vols (`Shifted_Black_vol_surface.xlsx`, sheet `3M_vols`; ICAP, 5 November 2025): 16 expiry tenors (1Y–30Y), 14 strikes (−1.5% to +10%), displacement $\delta = 3\%$. The surface has a pronounced negative skew at short tenors, reflecting persistent demand for negative-rate protection.

2.3.3 Credit Data

UniCredit EUR 5-year CDS (UNIC5YCDS, Investing.com, Dec 2015–Nov 2025) [Investing.com, 2025]: 38.78 bp on 5 November 2025. Recovery rate $R = 40\%$ (senior unsecured convention).

2.4 Term Structure Bootstrapping

2.4.1 Bootstrapping Methodology

Two-stage bootstrapping in QuantLib (TARGET, Modified Following): deposits Actual/360; IRS fixed-leg 30/360; zero rates Actual/365 continuously compounded.

Stage 1: Deposits. For each EURIBOR deposit with rate L and Actual/360 accrual fraction $\alpha(t_0, T)$:

$$P(t_0, T) = \frac{1}{1 + L \alpha(t_0, T)}. \quad (4)$$

The 1Y IRS overlaps with the 12M deposit and is excluded; swaps begin at 18M.

Stage 2: Swaps. For an IRS with par rate S_n , the floating leg PV collapses to $1 - P(t_0, T_n)$ under standard single-curve assumptions. With annual 30/360 accrual fractions $\alpha_i = \alpha(T_{i-1}, T_i)$:

$$P(t_0, T_n) = \frac{1 - S_n \sum_{i=1}^{n-1} \alpha_i P(t_0, T_i)}{1 + S_n \alpha_n}. \quad (5)$$

The 12M deposit provides $P(t_0, T_1)$ directly, so the swap bootstrap begins at 1Y without gap. Payment schedules are generated using QuantLib's `Schedule` object with annual frequency and Modified Following adjustment. A single-curve framework is used throughout; no tenor basis spread between 3-month and 6-month EURIBOR is applied. Results at selected IRS maturities are reported in Table 5.

Table 5: Bootstrapped discount factors and zero rates at selected maturities. Spot 7 Nov 2025. $z(T) = -\ln P/T$, ACT/365. *Nearest node to bond maturity (8.6Y).

Tenor	Maturity	Swap (%)	$P(t_0, T)$	Zero CC (%)
1Y	09 Nov 2026	2.1990	0.978074	2.2049
2Y	08 Nov 2027	2.1420	0.958460	2.1185
3Y	07 Nov 2028	2.1930	0.936980	2.1678
5Y	07 Nov 2030	2.3498	0.890091	2.3274
7Y	08 Nov 2032	2.4803	0.841574	2.4611
8Y*	07 Nov 2033	2.5398	0.817079	2.5235
10Y	07 Nov 2035	2.6330	0.769250	2.6220
15Y	07 Nov 2040	2.8330	0.652816	2.8410
20Y	07 Nov 2045	2.9040	0.557729	2.9174
25Y	07 Nov 2050	2.9090	0.482537	2.9129

The curve exactly reprices every quoted swap by construction. For any un-quoted point $T \in [T_k, T_{k+1})$, par rates are linearly interpolated:

$$S(T) = S(T_k) + \frac{T - T_k}{T_{k+1} - T_k} [S(T_{k+1}) - S(T_k)]. \quad (6)$$

Substituting into (5) gives an **explicit** procedure. A daily zero-rate grid is then generated and exported to `Interp_term_structure.xlsx`:

$$P(d) = \exp(-\hat{z}(d) \cdot d/365), \quad \hat{z}(d) = \text{interp}(d; \{d_k\}, \{z_k\}). \quad (7)$$

Linear interpolation was chosen for its simplicity and because it is sufficient for this instrument, whose cash flows are concentrated below 9Y where quoted instruments are dense.

2.5 Coupon Decomposition

Each future coupon uses (2): leveraged float at $1.6F_i$, long floorlet at 0%, short caplet at L^* , with

$$F_i = \frac{P(t_0, T_i^{\text{start}})/P(t_0, T_i^{\text{end}}) - 1}{\alpha_i^{\text{ACT}/360}}. \quad (8)$$

Periods 7–40 (34 options each side) priced via the DB model (Section 2.6); period 6 is already fixed.

2.6 Cap Strip Pricing

2.6.1 Model Selection

Standard Black (1976) is inadmissible (log-normal rates, zero probability to negative values). With EURIBOR $\approx 2\%$ and surface strikes down to -1.5% , we adopt the **Displaced Black (DB) model**

with $\delta = 3\%$ (VCAP3A EUR convention):

$$\text{Caplet}_i = \alpha_i P(t_0, T_i^{\text{pay}}) [(F_i + \delta)\Phi(d_1) - (L^* + \delta)\Phi(d_2)], \quad (9)$$

$$\text{Floorlet}_i = \alpha_i P(t_0, T_i^{\text{pay}}) [\delta\Phi(-d_2) - (F_i + \delta)\Phi(-d_1)], \quad (10)$$

$$d_1 = \frac{\ln[(F_i + \delta)/(K + \delta)] + \frac{1}{2}\sigma^2 T_i^{\text{reset}}}{\sigma\sqrt{T_i^{\text{reset}}}}, \quad d_2 = d_1 - \sigma\sqrt{T_i^{\text{reset}}}, \quad (11)$$

where α_i is ACT/360, T_i^{reset} is ACT/365, and Φ is the standard normal CDF.

2.6.2 Volatility Calibration

Bilinear interpolation at 8.60Y yields: $\sigma_{\text{cap}} = 8.2007\%$ ($K = 3.406\%$); $\sigma_{\text{floor}} = 11.8511\%$ ($K = 0\%$). One flat vol per strip is consistent with standard market practice for quoting cap and floor prices. The higher floor vol reflects the pronounced left skew of the EUR surface, where deep out-of-the-money floorlets command elevated implied vols due to persistent demand for negative-rate protection. Despite this, the large moneyness gap of 200–300 bp keeps individual floorlet values negligible. All 34 caplet d_1 values are strongly negative (from -8.83 at period 7 to -0.13 at period 40), confirming that every caplet is deeply out of the money and that caplet value is driven entirely by time value rather than intrinsic value. A $\sqrt{3/6}$ tenor adjustment applies beyond 2.5Y (3M vs 6M EURIBOR reference). The main limitation is the flat-vol assumption, which overprices near-term caplets relative to long-dated ones.

2.7 Credit Valuation Adjustment

Under independence between rates and default, $GP_D = GP_{\text{RF}} - \text{CVA}$:

$$\text{CVA} = (1 - R) N \sum_{i=1}^M P(0, T_i) [Q(T_{i-1}) - Q(T_i)], \quad (12)$$

where $N = \text{€}1,000$, $P(0, T_i)$ is the risk-free discount factor, and $Q(T_{i-1}) - Q(T_i)$ is the marginal default probability in $(T_{i-1}, T_i]$. Recovery is applied to the notional on default rather than to individual coupon cashflows, consistent with the bond's senior unsecured ranking. Flat hazard:

$$Q(0, T) = e^{-\lambda T}, \quad \lambda = s/(1 - R), \quad (13)$$

with s the CDS spread from UNIC5YCDS.

2.8 PCA-Based Risk Factor Model

2.8.1 Principal Component Analysis

The covariance matrix Σ of daily IRS changes (bp, 8 tenors) is decomposed as $\Sigma = \mathbf{A}\mathbf{D}\mathbf{A}^\top$. The covariance matrix is preferred over the correlation matrix because eigenvalues carry their natural interpretation as factor variances in bp^2 , directly usable for risk attribution without rescaling. Three factors are retained ($\geq 95\%$ cumulative variance); identities follow Litterman and Scheinkman [1991].

2.8.2 Linear Factor Model

$$\Delta \text{GP}(t) = \sum_{k=1}^3 \beta_k \Delta f_k(t) + \beta_{\text{cds}} \Delta \text{cds}(t) + \varepsilon(t), \quad (14)$$

where the IR factor betas are obtained by projecting the DV01 vector $\mathbf{d}$ onto eigenvector $\mathbf{a}_k$:

$$\beta_k = \mathbf{a}_k^\top \mathbf{d} = \sum_j a_{kj} \cdot \text{DV01}_j, \quad k = 1, 2, 3. \quad (15)$$

P&L variance:

$$\sigma_{\Delta \text{GP}}^2 = \sum_k \beta_k^2 \lambda_k + \beta_{\text{cds}}^2 \sigma_{\text{cds}}^2. \quad (16)$$

Exact 99% VaR/ES ($z_{0.99} = 2.3263$; Jorion 2007):

$$\text{VaR}_\alpha = z_\alpha \sigma_{\Delta\text{GP}}, \quad \text{ES}_\alpha = \frac{\phi(z_\alpha)}{1 - \alpha} \sigma_{\Delta\text{GP}}. \quad (17)$$

Euler component VaR: $\text{CVaR}_k = \beta_k^2 \lambda_k z_\alpha / \sigma_{\Delta\text{GP}}$ (sums to total VaR).

2.9 Hedging Strategy

2.9.1 Interest Rate Hedge

Zeroing portfolio factor betas: $\mathbf{A}^\top \mathbf{n} = -\beta^b$ [Tuckman and Serrat, 2013], where $\mathbf{A}$ is the 3×3 IRS factor-beta matrix (EUR/bp per EUR 1m notional). Three EUR receive-fixed IRS (1Y, 5Y, 10Y) are selected because they span the short, medium and long ends of the curve, providing near-orthogonal coverage of the level, slope and curvature dimensions.

2.9.2 Credit Hedge

$\beta_{\text{cds}} \approx [V_D(s+1\text{bp}) - V_D(s-1\text{bp})]/2$; CDS notional $N_{\text{CDS}} = -\beta_{\text{cds}}/\text{DV01}_{\text{CDS}}$, $\text{DV01}_{\text{CDS}} = A_{\text{risky}} \times 10^{-4}$, $A_{\text{risky}} = \sum_i \frac{1}{4} P(0, T_i) Q(0, T_i)$ [Hull, 2018].

2.10 Assumptions

1. IR factors and CDS spread are independent (no wrong-way risk).
2. Residuals ε are zero-mean i.i.d. Gaussian; excess kurtosis not modelled.
3. Bond DV01s use the risk-free function; credit does not affect IR sensitivities.
4. EUR 1m IRS minimum lot is binding.
5. CDS variance estimated Dec 2015–Nov 2025 (includes 2022–23 tightening).

3 Implementation

3.0.1 CVA Computation

Equation (12) over 36 intervals (periods 6–40 + maturity). $\lambda = 38.78\text{bp}/(10,000 \times 0.6) = 6.463 \times 10^{-4} \text{ yr}^{-1}$ (ACT/365). $GP_{\text{RF}} = \text{€}1,102.27$ from the cap-strip pricer. Per-period contributions saved to `q6.cva_detail.csv`.

3.0.2 Market-Implied CDS Spread

s^* solves $V^{\text{risky}}(s^*) = P_{\text{clean}}^{\text{mkt}} + \text{AI} = \text{€}1,012.879$. Root in $[0, 500]$ bp; Brent’s method converges to 0.01 bp in < 20 calls. Implied ≈ 130 bp (Section 4.1.6).

3.0.3 PCA, DV01 and Hedge Ratios

PCA: 2,576 daily IRS changes (1Y–20Y, 8 nodes; Dec 2015–Nov 2025) via `numpy.linalg.eigh`. DV01: each node bumped ± 1 bp; curve re-bootstrapped; $\text{DV01}_j = [GP_j^+ - GP_j^-]/2$ (vols fixed). IRS DV01s computed identically; hedge system solved by `numpy.linalg.lstsq`.

3.0.4 VaR, ES and Credit DV01

Credit DV01: ± 1 bp bump, full CVA loop. Exact VaR/ES from (17); MC cross-check: 10^6 Gaussian draws. Hedged betas: $\beta_k^{\text{port}} = \beta_k^{\text{bond}} + \sum_j H_{jk}$.

4 Results and Discussion

4.1 Part I: Bond Valuation

4.1.1 Bootstrapped Discount Curve

Figure 2 shows the bootstrapped EUR zero curve rising from 1.90% (O/N) to 2.91% at 25Y, then inverting mildly at the ultra-long end—consistent with gradual rate normalisation following ECB cuts from mid-2024. At bond maturity (8.6Y): $z = 2.53\%$, $P(0, T_{40}) = 0.8024$.

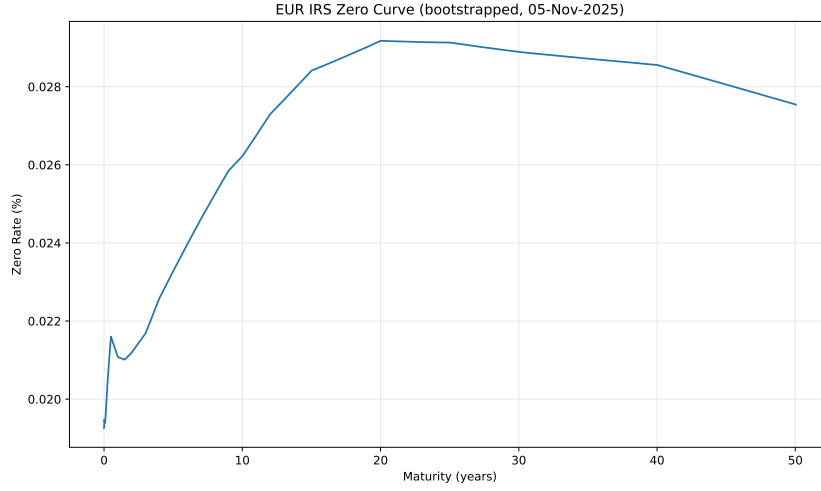


Figure 2: Bootstrapped EUR zero curve (c.c., ACT/365), 5 November 2025. EURIBOR deposits O/N–12M; IRS par rates 1Y–50Y.

4.1.2 Best and Worst Case Scenarios

Table 6: Best- and worst-case prices (EUR per €1,000 nominal, 5 November 2025).

Scenario	Assumption	Gross	Gross (%)	Clean
Worst	EURIBOR $\leq 0\%$ all resets	810.48	81.05	805.70
Best	EURIBOR $\geq 3.41\%$ all resets	1,227.78	122.78	1,223.00
Range		417.30	41.73	417.30

The fair value (EUR 1,072.98 risky clean) lies well inside this EUR 417 range, closer to the best-case bound.

4.1.3 Cap Strip Pricing

Table 7 reports the risk-free bond valuation decomposed into its four structural components.

Table 7: Risk-free bond valuation by component (EUR per EUR 1,000 nominal, trade date 5 November 2025). Displaced Black model, $\delta = 3\%$, $\sigma_{\text{cap}} = 8.2007\%$, $\sigma_{\text{floor}} = 11.8511\%$ at 8.60Y maturity.

Component	EUR	% of RF gross
PV(FRN, $1.6 \times L^{3M}$, periods 6–40)	+316.87	28.75
PV(floor strip, $K = 0\%$, periods 7–40)	+ 0.37	0.03
PV(cap strip, $K = 3.406\%$, periods 7–40)	– 17.35	–1.57
PV(notional, at T_{40})	+802.38	72.79
RF Gross Price	1,102.27	100.00

The cap strip costs EUR 17.35 (1.57% of RF gross), entirely from time value: all 34 caplets are out of the money at trade date, with forward rates ranging from 2.12% to 3.04% against a breakeven of $L^* = 3.406\%$. The floor strip contributes only EUR 0.37 (0.03%) because EURIBOR is well above zero across the entire forward curve; the Displaced Black model assigns positive probability to negative rates under the 300 bp displacement, producing a small but non-zero floor value. The notional dominates at 72.79%, consistent with the bond’s approximately 8.6-year remaining life and a discount factor of $P(0, T_{40}) = 0.8024$.

4.1.4 CVA and Risky Bond Price

CDS 38.78 bp, $R = 40\%$: $\lambda = 6.463 \times 10^{-4} \text{ yr}^{-1}$; cumulative default probability 5.41% over 8.6Y; CVA = EUR 29.29 (entirely on coupon intervals). Table 8 summarises the full valuation.

Table 8: Bond valuation summary (EUR per €1,000, 5 Nov 2025; CVA at 38.78 bp CDS, $R = 40\%$).

Component	EUR	% of RF gross
PV(FRN, $1.6 \times L^{3M}$)	316.87	28.75
PV(floor strip, $K = 0\%$)	0.37	0.03
PV(cap strip, $K = 3.406\%$)	-17.35	-1.57
PV(notional)	802.38	72.79
RF Gross	1,102.27	100.00
CVA	29.29	2.66
Risky Gross	1,072.98	97.34
Accrued Interest	4.78	
Risky Clean	1,068.20	

CVA of EUR 29.29 (2.66%) is a modest adjustment consistent with UniCredit's A2 credit rating. The floor contributes only EUR 0.37 (EURIBOR well above zero along the forward curve); the cap costs EUR 17.35 (1.57%), reflecting the bond's embedded short caplet position as forwards approach the 3.41% effective strike.

4.1.5 Pricing Summary and Expected Cash Flows

Table 8 presents the full bond valuation decomposition and Table 9 reports the expected coupon by period. Figure 3 plots the complete cash flow profile from period 6 to period 40. All 34 remaining caplets are OTM (forwards 2.12%–3.04%, below $L^* = 3.406\%$). Expected coupons rise from €8.47 (period 7) to €12.16 (period 40), tracking the forward curve.

Table 9: Expected coupons (selected). DB model, $\sigma_{\text{cap}} = 8.20\%$, $\sigma_{\text{floor}} = 11.85\%$, $\delta = 3\%$. Per €1,000 nominal.

Period	Payment	Fwd EURIBOR	Exp. coupon (€)	Caplet PV (€)	Floorlet PV (€)
6	12 Dec 2025	2.029% (fixed)	8.116	—	—
7	12 Mar 2026	2.118%	8.47	0.000	0.000
15	13 Mar 2028	2.215%	8.86	0.045	0.000
25	12 Sep 2030	2.609%	10.44	0.499	0.009
35	14 Mar 2033	2.898%	11.59	1.074	0.026
40	12 Jun 2034	3.040%	12.16	1.369	0.035
Total (periods 7–40)			316.870	17.346	0.369

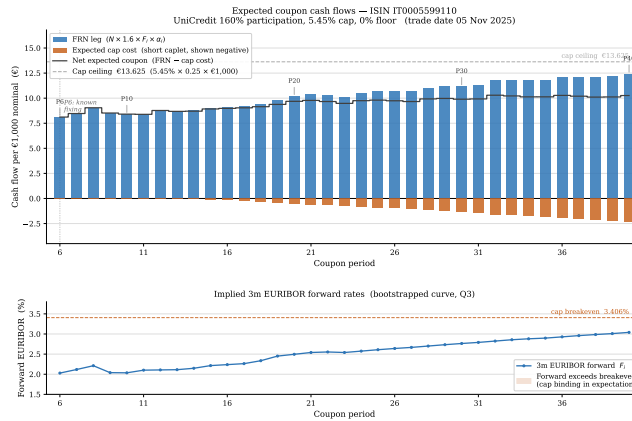


Figure 3: Expected coupon cash flows per €1,000, periods 6–40. All below the €13.625 cap (dashed); rising profile tracks the forward curve.

4.1.6 Market Price Comparison and Implied CDS Spread

EuroTLX clean price: **100.81%** (€1,008.10; Borsa Italiana, 5 Nov 2025). Tables 10–11 compare model and market values. The €60 gap vs risky fair value exceeds the EUR 29.29 CVA, indicating the

Table 10: Clean price comparison. Per €1,000 nominal, 05 Nov 2025.

Price measure	Clean (€)	% of par
RF fair value (DB model)	1,097.49	109.75%
Risky fair value (CVA-adjusted)	1,068.20	106.82%
EuroTLX market price	1,008.10	100.81%
Market vs RF	−89.39	−8.94 pp
Market vs risky	−60.10	−6.01 pp

market prices in a materially higher credit premium. The implied spread (≈ 130 bp, Brent) exceeds 38.78 bp due to: (i) maturity mismatch (5Y CDS vs 8.6Y bond, upward credit curve); (ii) bond-CDS basis (cash bonds trade cheap for liquidity and balance-sheet reasons); (iii) distribution costs (1.69% issuance commission persists); (iv) model risk (flat-vol overestimates RF price).

Table 11: Model vs market summary. 05 Nov 2025.

Metric	Value	Source
RF clean price	109.75%	DB model
Risky clean price	106.82%	CVA-adjusted
EuroTLX clean price	100.81%	Borsa Italiana
5Y CDS spread	38.78 bp	investing.com
Implied CDS spread	≈ 130 bp	Brent root-finding
Bond-CDS basis	≈ -91 bp	bond cheap vs CDS

4.2 Part II: Risk Measurement and Hedging

4.2.1 Yield Curve Factors and Bond Sensitivity

PCA on 2,576 daily EUR IRS changes (Dec 2015–Nov 2025) yields three factors explaining 97.6% of variance (Table 12). **PC1 (Level)** ($\lambda_1 = 94.72$ bp², 86.4%): uniformly positive loadings (0.171 at 1Y, plateau ≈ 0.40 at 5Y–10Y; suppressed 1Y reflects ECB anchoring). **PC2 (Slope)** ($\lambda_2 = 10.46$ bp², 9.5%): negative 1Y–5Y, crosses zero near 7Y, +0.48 at 20Y (bull/bear steepener). **PC3 (Curvature)** ($\lambda_3 = 1.76$ bp², 1.6%): 1Y peak (0.66), −0.46 belly at 5Y, butterfly [Litterman and Scheinkman, 1991].

Table 12: Eigenvalue decomposition (8 tenors, Dec 2015–Nov 2025).

Factor	λ (bp ²)	$\sqrt{\lambda}$ (bp/day)	Expl. (%)	Cum. (%)
PC1 Level	94.72	9.73	86.4	86.4
PC2 Slope	10.46	3.23	9.5	95.9
PC3 Curvature	1.76	1.33	1.6	97.6
PC4–PC8	2.67	—	2.4	100.0

Table 13: Normalised eigenvector loadings (EUR IRS, 8 tenors).

Tenor	PC1 (Level)	PC2 (Slope)	PC3 (Curvature)
1Y	0.171	−0.356	0.661
2Y	0.310	−0.500	0.245
3Y	0.353	−0.400	−0.151
5Y	0.394	−0.174	−0.461
7Y	0.401	0.030	−0.327
10Y	0.397	0.233	−0.055
15Y	0.381	0.389	0.192
20Y	0.363	0.476	0.347

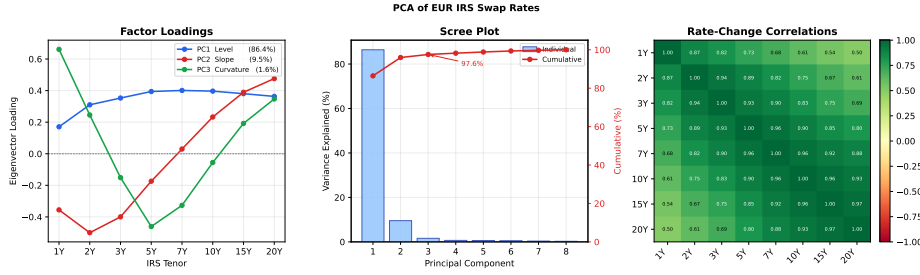


Figure 4: PC1–PC3 loadings. PC1 uniformly positive (level); PC2 crosses zero near 7Y (slope); PC3 negative belly at 5Y (curvature).

PC1 score volatility rose sharply during the ECB tightening cycle (February 2022 – September 2023), peaking at 17.7 bp/day, before declining as the ECB began cutting rates in June 2024. The bond’s IR DV01 is +0.137 EUR/bp ($\sim 340\times$ smaller than a comparable fixed-rate bond), reflecting the par FRN theorem: the $1.6\times$ floating coupon provides natural IR immunisation, with the residual arising from the 30/360 vs ACT/360 convention mismatch and the embedded cap and floor. Table 14 reports the full DV01 vector alongside the $\pm 1\sigma$ bond price sensitivity to each PCA factor shock. Bond factor betas are obtained by projecting the DV01 vector onto the eigenvectors via (15): $\beta_l = +0.0531$, $\beta_s = -0.0520$, $\beta_c = -0.0609$ EUR/bp. A $\pm 1\sigma$ level shock causes a price change of $\pm$ EUR 0.52; slope and curvature shocks produce $\mp$ EUR 0.17 and $\mp$ EUR 0.08 respectively.

Table 14: Bond DV01 and factor sensitivities (EUR per €1,000, vols fixed).

Tenor	DV01 (EUR/bp)			
1Y	−0.0140	Factor	σ	$+1\sigma$
2Y	+0.0156			-1σ
3Y	+0.0637			
5Y	+0.0876			
7Y	+0.0233			
10Y	−0.0395			
Total	+0.1365	PC1 Level	9.73	+0.52
		PC2 Slope	3.23	−0.17
		PC3 Curvature	1.33	−0.08

4.2.2 Interest Rate Hedging

Hedge notionals: +EUR 267 (IRS 1Y, rec.-fixed), +EUR 376 (IRS 5Y, rec.-fixed), −EUR 60 (IRS 10Y, pay-fixed) per €1,000 bond. The condition number of the hedge matrix is 6.3 (excellent), and the hedge achieves 100% effectiveness (machine-precision factor-beta neutralisation). At the full issue scale of EUR 20m (20,000 bonds), the notionals round to EUR 5m (IRS 1Y), EUR 8m (IRS 5Y) and EUR −1m (IRS 10Y) at the EUR 1m minimum lot, with all rounded values exceeding the minimum trade size. Practical limitations: (i) *Convexity*: the linear hedge under-performs for large moves due to the cap’s negative convexity. (ii) *Basis risk*: the bond references 3M EURIBOR while the IRS references 6M; a 3M/6M basis swap would eliminate this residual. (iii) *Dynamic betas*: loadings drift

as rates move and the bond ages, requiring periodic rebalancing. (iv) *Minimum lot*: portfolios below $\approx$ EUR 5m cannot implement the hedge without material deviation.

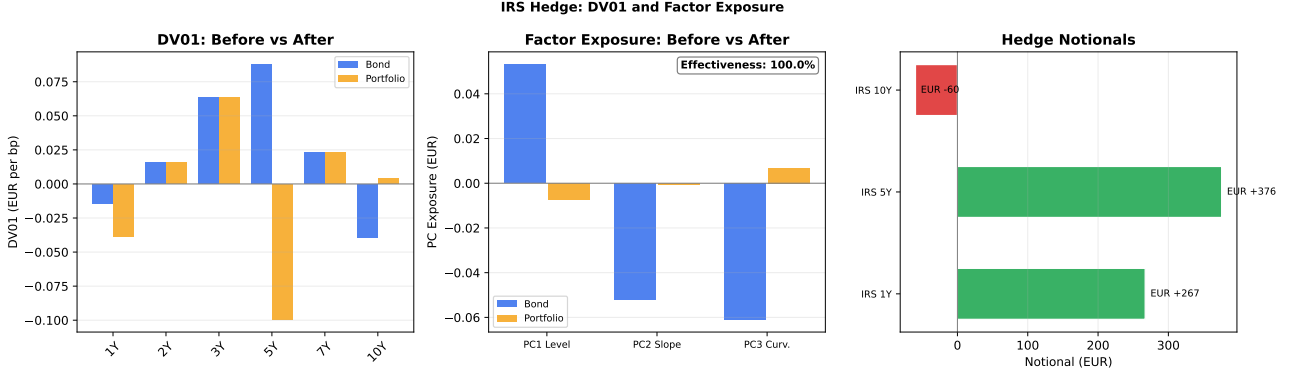


Figure 5: IR hedge: DV01 and factor exposure before/after. Left: DV01 by tenor. Centre: factor betas (post-hedge at machine precision). Right: rounded hedge notionals.

4.2.3 Credit Risk and CDS Hedge

$\beta_{\text{cds}} = -0.7352$ EUR per +1 bp CDS widening per €1,000. The relationship is approximately linear within ± 20 bp of the current spread; beyond that, positive credit convexity causes the DV01 to become less negative (bond value falls at a decreasing rate as spreads widen). The CDS protection hedge notional is EUR 1,596 per EUR 1,000 bond (buy protection, 5Y quarterly-premium CDS). This is computed from a risky annuity of $A_{\text{risky}} = 4.609$, giving $\text{DV01}_{\text{CDS}} = 4.609 \times 10^{-4}$ EUR/bp per EUR 1 notional. Practical limitations: maturity mismatch (5Y CDS vs 8.6Y bond); static hedge ratio (the risky annuity changes over time); CDS/bond basis divergence.

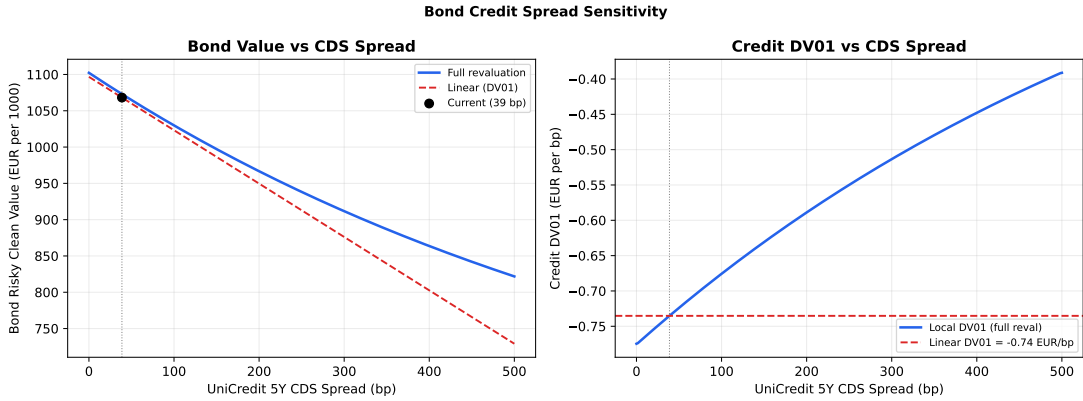


Figure 6: Credit DV01. Left: risky clean price vs CDS spread (full revaluation and linear approximation at 38.78 bp). Right: local DV01 (positive convexity; linear accurate within ± 20 bp).

4.2.4 Factor Model, VaR and Decomposition

Factor model. Using the betas from the preceding analysis and the historical CDS variance $\sigma_{\text{cds}}^2 = 21.82 \text{ bp}^2$ ($\sigma = 4.67 \text{ bp/day}$) from 2,564 daily observations of UNIC5YCDS, substituting into (16) gives $\sigma_{\Delta \text{GP}} = 3.482 \text{ EUR/day}$. The CDS term contributes $(-0.7352)^2 \times 21.82 = 11.79 \text{ EUR}^2$ out of a total variance of 12.09 EUR^2 , confirming that credit spread risk overwhelmingly dominates interest rate risk in this bond. **VaR and ES.** Applying (17) with $\sigma_{\Delta \text{GP}} = 3.482 \text{ EUR}$ gives the results in Table 15. The Monte Carlo and analytical results agree to within 0.3%, validating the Gaussian-linear model specification (Figure 7).

Table 15: VaR/ES at 99%, 1-day (EUR per €1,000).

Metric	Unhedged	Hedged
$\sigma_{\Delta GP}$	3.48	3.43
VaR _{99%} exact	8.09	7.99
ES _{99%} exact	9.27	9.15
VaR _{99%} MC	8.07	7.99
ES _{99%} MC	9.24	9.15

Model limitations: excess kurtosis; linear approximation ignores cap/floor convexity; constant betas ignore drift.

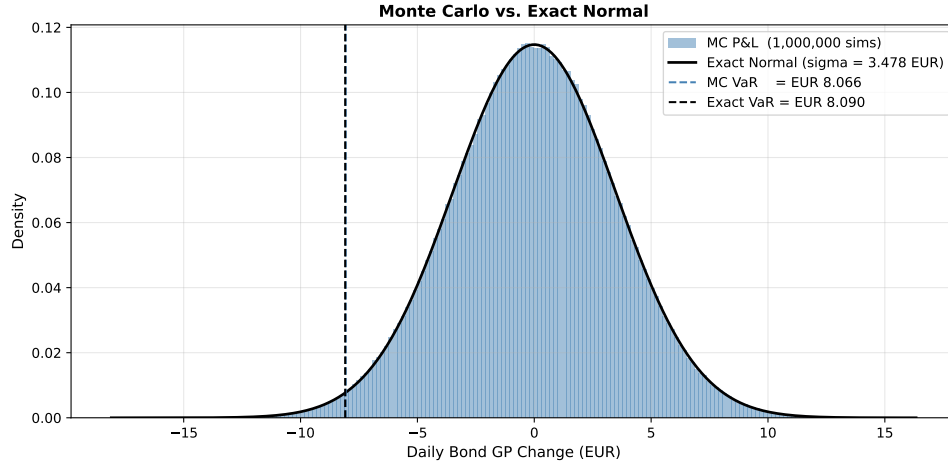


Figure 7: MC P&L (10^6 sims) with Gaussian overlay. Dashed: 99% VaR at EUR 8.07.

IRS hedge reduces VaR by only 1.2% (EUR 8.09 $\rightarrow$ EUR 7.99). The portfolio residual VaR is driven entirely by the unhedged CDS exposure, confirming that this bond is, in economic substance, a credit instrument. Table 16 shows the full decomposition.

Table 16: Euler component VaR (99%, 1-day; EUR per €1,000).

Factor	Unhedged		Hedged	
	CVaR	%	CVaR	%
Level (PC1)	0.179	2.2	≈ 0	—
Slope (PC2)	0.019	0.2	≈ 0	—
Curvature (PC3)	0.004	0.05	≈ 0	—
CDS spread	7.888	97.5	7.888	100
Total	8.09	100	7.99	100

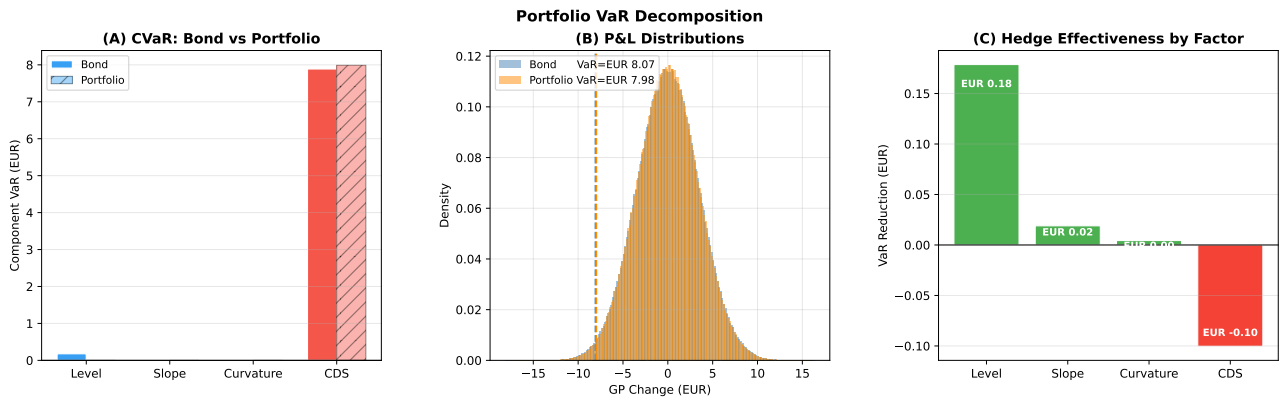


Figure 8: VaR decomposition and hedge effectiveness.

Table 17 consolidates fair values and sensitivities.

Table 17: Fair values and factor sensitivities per €1,000 (5 Nov 2025).

Instrument	GP	β_l	β_s	β_c	β_{cfs}
UniCredit bond 2034	1,072.98	+0.0531	-0.0520	-0.0609	-0.7352
IRS 1Y (rec.-fixed)	0.00	-0.0045	+0.0093	-0.0173	—
IRS 5Y (rec.-fixed)	0.00	-0.0693	+0.0305	+0.0810	—
IRS 10Y (pay-fixed)	0.00	+0.0207	+0.0122	-0.0029	—
Portfolio	1,072.98	≈ 0	≈ 0	≈ 0	-0.7352

5 Conclusions

Valuation. RF gross EUR 1,102.27; CVA EUR 29.29 at 38.78 bp CDS ($R = 40\%$); risky clean EUR 1,068.20. **Interest rate risk.** IR DV01 +0.137 EUR/bp ($\sim 340\times$ below a fixed-rate equivalent). A perfect three-IRS hedge reduces VaR by only 1.2%; IR risk is 2.5% of daily variance. **Credit risk dominates.** CDS accounts for 97.5% of VaR. A 100 bp CDS widening reduces fair value by $\approx$ EUR 73.5 (6.9%). Investors should view this bond as a leveraged credit position with an embedded rate cap. A full CDS hedge (EUR 1,596 notional) eliminates the dominant risk but also most of the carry. **Investor recommendation.** Suitable for investors with a positive UniCredit credit view expecting EURIBOR above $\approx 1\%$: the $1.6\times$ participation and 5.45% cap are attractive in the current environment. The 0% floor is unlikely to activate without a return to negative-rate policy. **Limitations.** Gaussian model understates tail risk (excess kurtosis; jump events in CDS). Credit-rate independence breaks down in flight-to-quality episodes. Static hedge ratios require periodic rebalancing. Extensions: jump-diffusion CDS, joint IR/credit model, dynamic cap delta-hedging.

References

- John C. Hull. *Options, Futures, and Other Derivatives*. Pearson, 10th edition, 2018.
- Investing.com. UniCredit CDS 5 year EUR (UNIC5YEUAM=R) historical data. <https://www.investing.com/rates-bonds/unicredit-cds-5-year-eur-historical-data>, 2025.
- Philippe Jorion. *Value at Risk: The New Benchmark for Managing Financial Risk*. McGraw-Hill, 3rd edition, 2007.
- Robert Litterman and José Scheinkman. Common factors affecting bond returns. *Journal of Fixed Income*, 1(1):54–61, 1991.
- Suomen Pankki (Bank of Finland). EURIBOR interest rate statistics. <https://www.suomenpankki.fi/en/Statistics/interest-rates/>, 2025.

Bruce Tuckman and Angel Serrat. *Fixed Income Securities: Tools for Today's Markets*. Wiley Finance, 3rd edition, 2013. Chapters 18 and 21.

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